

0.1 Pseudopotential Kohn–Sham accuracy

We next assess the accuracy of SPARC-atomSFE for pseudopotential Kohn–Sham DFT calculations through comparisons with the literature. In particular, we compare against the spectral scheme based SPARC-atom code, implemented as part of the M-SPARC package, while considering four exchange-correlation functionals: LDA, PBE, rSCAN, and PBE0. We compare the results for seven atoms spanning a range of chemical environments—He, N, O, Fe, Mn, Mo, and Cs. The distribution of the differences in the results is shown as violin plots in Figure 1. We observe that total energies and occupied valence eigenvalues agree to 10^{-6} Ha or better for LDA, PBE, and rSCAN. The exception is PBE0, which exhibits errors approximately an order of magnitude larger; this can be attributed to the fact that the quadrature in the spectral scheme used by SPARC-atom is limited to the degree of the polynomial used, making it difficult to converge calculations involving exact exchange, where higher-order quadrature is required. This limitation can be overcome by adopting a differential approach for the exact exchange operator, as done in the present work. These results demonstrate the accuracy of SPARC-atomSFE for atomic structure calculations based on pseudopotential Kohn–Sham DFT.

As supplementary material to the accuracy discussion above, we tabulate the full numerical data underlying Figure 1 in Tables 1, 2, 3, and 4 (LDA-SVWN, GGA-PBE, rSCAN, and PBE0, respectively).

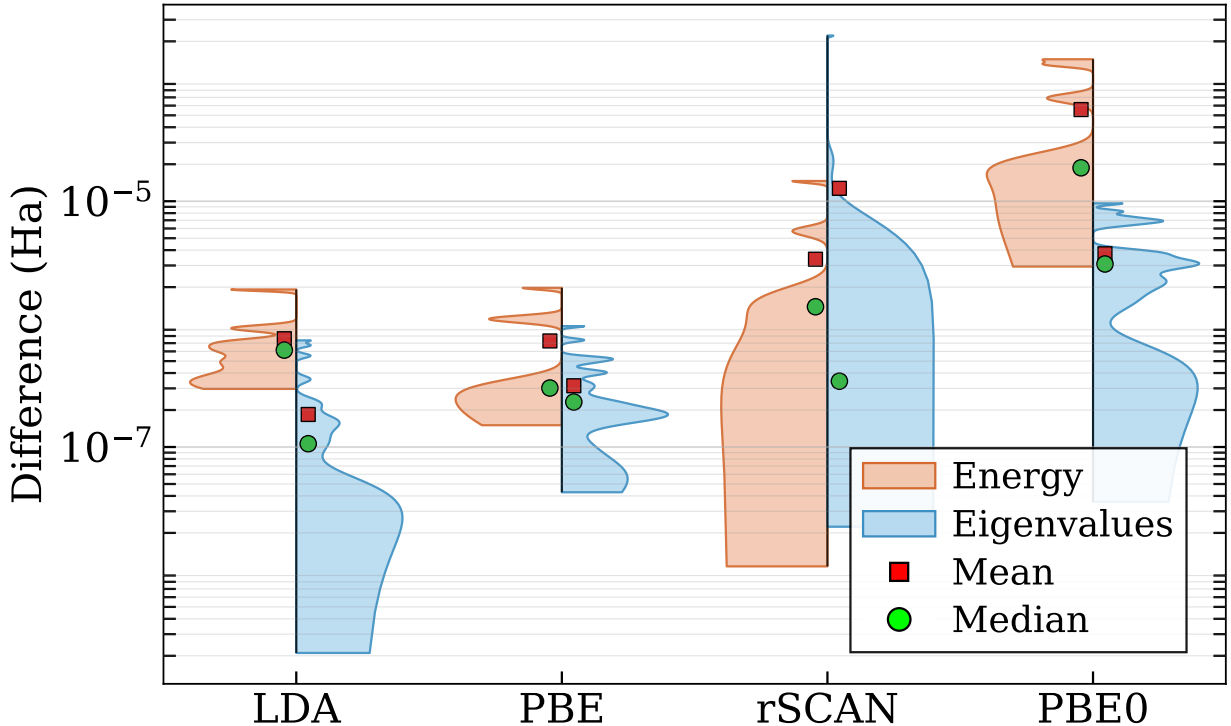


Figure 1: Comparison of SPARC-atomSFE pseudopotential Kohn–Sham DFT results with the M-SPARC SPARC-atom reference for various exchange-correlation functionals. The energy and eigenvalue errors denote the absolute differences in the corresponding quantities.

Table 1: Comparison of the total energies and eigenvalues for the LDA-SVWN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.800084291	2.800083994	2.97×10^{-7}	1	0	2	0.57046	0.57046	1.52×10^{-7}
N	7	10.0228671	10.02286518	1.91×10^{-6}	2	0	2	0.67708	0.677081	6.72×10^{-7}
					2	1	3	0.266124	0.266124	5.53×10^{-7}
O	8	16.21748133	16.21748084	4.91×10^{-7}	2	0	2	0.873064	0.873065	7.37×10^{-7}
					2	1	4	0.338133	0.338133	3.56×10^{-7}
Mn	25	105.2337256	105.2337263	7.14×10^{-7}	3	0	2	3.138755	3.138755	4.54×10^{-8}
					3	1	6	2.002629	2.002629	3.21×10^{-9}
					3	2	5	0.257732	0.257732	2.12×10^{-7}
					4	0	2	0.194097	0.194097	1.39×10^{-8}
Fe	26	116.7411744	116.741174	3.69×10^{-7}	3	0	2	3.436694	3.436694	3.18×10^{-8}
					3	1	6	2.202574	2.202574	2.82×10^{-8}
					3	2	6	0.285108	0.285108	2.5×10^{-8}
					4	0	2	0.201428	0.201428	3.84×10^{-8}
Mo	42	69.69080029	69.69080091	6.14×10^{-7}	4	0	2	2.367637	2.367637	1.04×10^{-7}
					4	1	6	1.418245	1.418244	1.7×10^{-7}
					4	2	5	0.144562	0.144562	1.09×10^{-7}
					5	0	1	0.158758	0.158758	3.55×10^{-8}
Cs	55	23.31026826	23.31026919	9.25×10^{-7}	5	0	2	0.988689	0.988689	2.37×10^{-7}
					5	1	6	0.500169	0.500169	1.55×10^{-7}
					6	0	1	0.081715	0.081715	2.11×10^{-9}

Table 2: Comparison of the total energies and eigenvalues for the GGA-PBE exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.853330662	2.853330926	2.64×10^{-7}	1	0	2	0.579311	0.579311	2.35×10^{-7}
N	7	10.05930315	10.05930012	3.03×10^{-6}	2	0	2	0.682914	0.682914	3.92×10^{-7}
					2	1	3	0.260551	0.260551	6.63×10^{-7}
O	8	16.27370467	16.27370633	1.66×10^{-6}	2	0	2	0.880576	0.880576	6.52×10^{-7}
					2	1	4	0.331872	0.331872	2.07×10^{-7}
Mn	25	105.3660140	105.3660158	1.83×10^{-6}	3	0	2	3.156010	3.156010	1.38×10^{-7}
					3	1	6	2.006265	2.006265	9.34×10^{-8}
					3	2	5	0.248834	0.248834	3.64×10^{-7}
					4	0	2	0.187686	0.187687	4.18×10^{-7}
Fe	26	116.7788402	116.7788403	9.29×10^{-8}	3	0	2	3.455077	3.455077	1.76×10^{-9}
					3	1	6	2.206535	2.206535	8.20×10^{-8}
					3	2	6	0.275801	0.275801	3.47×10^{-9}
					4	0	2	0.194482	0.194482	2.37×10^{-8}
Mo	42	69.67943092	69.67942912	1.79×10^{-6}	4	0	2	2.364713	2.364713	1.15×10^{-7}
					4	1	6	1.414321	1.414321	1.21×10^{-8}
					4	2	5	0.137922	0.137922	1.82×10^{-7}
					5	0	1	0.150181	0.150181	4.06×10^{-8}
Cs	55	23.24984140	23.24984231	9.15×10^{-7}	5	0	2	0.982389	0.982389	7.55×10^{-8}
					5	1	6	0.496788	0.496788	1.59×10^{-7}
					6	0	1	0.076670	0.076669	3.29×10^{-8}

Table 3: Comparison of the total energies and eigenvalues for the rSCAN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.803317365	2.803317354	1.07×10^{-8}	1	0	2	0.605361	0.605361	6.61×10^{-8}
N	7	9.620390181	9.620390518	3.37×10^{-7}	2	0	2	0.729363	0.729363	1.45×10^{-7}
					2	1	3	0.26564	0.265639	3.03×10^{-7}
O	8	15.70167901	15.70168048	1.47×10^{-6}	2	0	2	0.940105	0.940105	5.79×10^{-7}
					2	1	4	0.343355	0.343355	4.7×10^{-8}
Mn	25	99.17611481	99.17611343	1.38×10^{-6}	3	0	2	3.279161	3.279161	3.62×10^{-7}
					3	1	6	2.08522	2.085219	3.09×10^{-7}
					3	2	5	0.251825	0.251825	3.86×10^{-7}
					4	0	2	0.195917	0.195917	2.25×10^{-8}
Fe	26	118.0039187	118.003913	5.71×10^{-6}	3	0	2	3.585989	3.585987	2.53×10^{-6}
					3	1	6	2.293534	2.293532	2.09×10^{-6}
					3	2	6	0.281913	0.281911	1.65×10^{-6}
					4	0	2	0.203008	0.203008	6.33×10^{-7}
Mo	42	68.16700484	68.16700474	9.92×10^{-8}	4	0	2	2.45666	2.456659	3.86×10^{-7}
					4	1	6	1.469806	1.469806	3.24×10^{-7}
					4	2	5	0.13904	0.13904	2.88×10^{-7}
					5	0	1	0.154164	0.154164	7.69×10^{-8}
Cs	55	20.17964157	20.17965618	1.46×10^{-5}	5	0	2	1.023347	1.023124	2.23×10^{-4}
					5	1	6	0.51887	0.51887	1.72×10^{-7}
					6	0	1	0.076458	0.076437	2.12×10^{-5}

Table 4: Comparison of the total energies and eigenvalues for the PBE0 exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			n	ℓ	g_{nl}	$-\varepsilon_{nl}$ (Ha)		
		atomSFE	M-SPARC	Difference				atomSFE	M-SPARC	Difference
He	2	2.858444072	2.858447011	2.94×10^{-6}	1	0	2	0.668803	0.668805	1.88×10^{-6}
N	7	9.8409288	9.840936922	8.12×10^{-6}	2	0	2	0.781191	0.781194	3.11×10^{-6}
					2	1	3	0.268917	0.268919	1.45×10^{-6}
O	8	16.01035357	16.01037228	1.87×10^{-5}	2	0	2	0.999295	0.999299	3.78×10^{-6}
					2	1	4	0.36414	0.364142	2.37×10^{-6}
Mn	25	103.7537001	103.7538296	1.3×10^{-4}	3	0	2	3.363098	3.363104	6.43×10^{-6}
					3	1	6	2.162649	2.162657	8.26×10^{-6}
					3	2	5	0.258815	0.258822	6.89×10^{-6}
					4	0	2	0.214786	0.214786	4.08×10^{-7}
Fe	26	115.7186491	115.718792	1.43×10^{-4}	3	0	2	3.665564	3.665571	6.96×10^{-6}
					3	1	6	2.363997	2.364006	9.61×10^{-6}
					3	2	6	0.298827	0.298834	7.47×10^{-6}
					4	0	2	0.222093	0.222093	3.07×10^{-7}
Mo	42	68.96799747	68.96806647	6.9×10^{-5}	4	0	2	2.526861	2.526864	3.36×10^{-6}
					4	1	6	1.53111	1.531114	3.84×10^{-6}
					4	2	5	0.144336	0.144338	2.22×10^{-6}
					5	0	1	0.147808	0.147807	3.82×10^{-7}
Cs	55	22.42005527	22.42007389	1.86×10^{-5}	5	0	2	1.066609	1.066612	2.9×10^{-6}
					5	1	6	0.551291	0.551294	3.06×10^{-6}
					6	0	1	0.076059	0.076059	3.59×10^{-8}